

CAP 6635: Artificial Intelligence (2026 Spring)

Research Project Report Instruction

(For Students Substituting Final Exam) (19 Points)

Due date [May 3 2026, FIRM]

This instruction only applies to students who intend to use a research project report to substitute the final exam. If you DO wish to participate in the final exam, please take the exam on May 6, 2026.

- Student cannot participate in final exam and turn in a “Research Report”! A -5 pt penalty will be applied to such submissions.
- Research reports can NOT be a literature review of existing work in the field. A literature review research report will receive grade 0.
- For team projects (i.e., involving more than one member), only team captain will submit the report for grading. A -5 pt penalty will be added to a team project, if more than one report was submitted from a team.

The grading of the term project report is based on the following criteria.

1. **Overall [3 pts]:** You should organize your report in IEEE format, with 4,000 words minimum. **Tables/figures do not count towards the word limitation.** Your report must use IEEE word or Latex template from the following URL
 - a. **Template:** http://www.ieee.org/conferences_events/conferences/publishing/templates.html.
 - b. **Plagiarism:** You cannot copy any sentences, paragraphs, or figures, from any external sources (such as published papers or Internet). If Turnitin indicates that a submission is over 30% similar to any other submissions, the instructor will carry out a Plagiarism investigation.
 - c. If you must cite a figure/graph published somewhere else, please properly cite the source of the reference [**F grade if plagiarism check returns over 50% similarity to any priori work**].
[Grading of number of words, content organization, grammars, and typos are included in the “Overall”]
2. **Title and Abstract [1 pt]:** Your report should have a brief and informative title and an abstract. The abstract should have 200-300 words, which summarizes the problem you intend to address in the report (e.g., zero-shot LLM text classification, transfer learning using naïve bayes classification etc.). Briefly describe designs and solutions which will be proposed in the report, and briefly summarize any conclusions the report intends to draw. [**200-300 words**]
3. **Introduction [2 pts]:** Your report should have an introduction section with 500 – 1000 words. The introduction should clearly state
 - a. what is the research problem to be studied in the report (**0.5 pt**);

- b. the motivation of the problem studied in your report (0.5 pt);
- c. how are the problem solved by existing methods, if any (0.5 pt);
- d. a brief description about the method you will propose in the report (0.5 pt).

4. **Related Work [2 pts]:** Your report should have a related work section to summarize works (algorithms) which already exist to solve the problem. For example, if you are trying to address transfer learning for naïve bayes classification using small number of training samples, you need to discuss how transfer learning was used in naïve bayes classification, what are existing methods for short message classification, what are common approach to transfer knowledge from a source, such as email spam, to a target domain, i.e., the short message classification. etc. [500-1000 words]
 - a. The related work section of your report must cite at least 10 peer reviewed references.
5. **Main body [5 pts]:** In the body of your project, you will need to provide technical details of your design [2000-2500 words]
 - a. If you are proposing a new approach to transfer a naïve bayes classifier train from one dataset to be used to another dataset, you will need to describe your designs. Describe the motivation of your design, and why do you think this would work to solve the problems [2 pts]
 - b. Use flowcharts, figures, or some pseudo-code to describe your algorithm details. [Use at least two figures (or flowcharts) to demonstrate the system framework or architecture]. Source code cannot be included in the report to form part of the report [2 pts]
 - c. Explain how your design is different from existing work [1 pt]
 - d. If your report is about experimental studies, you will need to provide a brief description about your learning/classification methods, the benchmark datasets, and different measures applied. You should also explain how the experiments are carried out in your study, and what type of empirical study goals you intend to achieve.
6. **Experiments [4 pts]:** In the experiments, you need to introduce (1) main purpose of the experimental studies; (2) what are the tools used to design the algorithms; (3) what are the baseline methods for comparisons; and (4) what are the performance measures and data used for empirical studies. You should also use figures and tables to report the results collected from your studies, and summarize the experimental results [1000-1500 words].
 - a. Experimental settings: including an introduction of baseline methods, programming tools/languages, the setting of the parameters used for different methods. [1 pt]
 - b. Benchmark data: Provide detailed description about data used for your study, including detailed information about the size/dimension of the data.[1 pt]
 - c. Baseline methods: In order to demonstrate the performance of your method, you will need to use a baseline approach, and compare the performance of your design with the baseline. For example, a simple naïve bayes classifier trained from 5% of the target dataset (i.e., the SMS dataset) can be used as a baseline to demonstrate that if a classifier is trained from a small

target set it might not perform very well. In this way, the experiments will demonstrate the merits of your method, and validate your hypothesis [1 pt]

- d. The results: The detailed results reported in figures/tables with necessary analysis and descriptions. You will need to include at least one figure and one table to show the results. [1 pts]
 - e. Analysis of the Results: Please compare the performance of your method and the baseline approach, and analyze why your method can obtain a good performance. Please also add a case study example (e.g. an example of a review report and the predicted result from your method) to why this sample was predicted with the reported score [1 pt]
7. **Conclusions [1 pt]:** In the conclusions, you should briefly summarize the research problem studied in your report. Explain what you have done and summarize the major findings. Draw any informative conclusions, which can be useful to guide the followers [150-200 words]
8. **References [1 pts]:** Please cite at least 12 relevant references in your report. At least 5 of them must be from 2019 or after.

Plagiarism Check

All reports are submitted through Turnitin.

Reports receiving 50% similarity scores to other sources will receive an F grade.

Reports receiving 30% - 50% similarity scores to other sources will receive a penalty score calculated using following formula:

$$-(\text{Your similarity score} - 0.3) * 100$$

For example, if the similarity score is 40%, the penalty score is

$$-(0.4 - 0.3) * 100 = -10$$